Stat 236 Final Project Proposal

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Topic: Determining Polarization in Education Politics Through Text and Network Analysis of School Board Meetings

Description: *Has local education politics become more partisan?* Political polarization in various areas of U.S. local politics has increased significantly in the United States (U.S.), deepening ideological and social divides across political groups (Iyengar 2019). These phenomena have significant effects for K-12 education in the U.S., shaping what is taught, who gets to learn, and which policies are implemented in classrooms. The COVID-19 pandemic exacerbated these challenges, transforming schooling into a national battlegrounds over topics such as school closures, masking, and later, a flashpoint in a broader culture war, introducing gag orders, (Hartney and Finger, 2022). Understanding how polarization has changed within the education sector can help policymakers re-assess the policies that divide communities, and address diverse perspectives when developing education policy. As such, for our final project, we are interested in studying how polarization in education politics has evolved over time, using data from school board meeting transcripts across the U.S. starting from 2005.

We will use text-as-data methods to examine changes in the language, topics, and political alignment of school board meetings before, during, and after the pandemic (Grimmer & Stewart, 2013). Specifically, we plan to estimate topic prevalence over time using topic models such as LDA or structural topic models (Roberts et al., 2014), and to measure partisanship in language using approaches developed for congressional speech (Gentzkow et al., 2019). To understand what drives differences between pre- and post-pandemic discourse, we will identify the linguistic features and topics most associated with the shift. Once we have district-level

polarization measures, we will use network analysis to examine how topics and language are connected across districts, including community detection (Newman, 2006), to test whether polarization is widespread or concentrated in particular local areas based on their partisanship or geography. If time allows, we aim to estimate the effects of COVID-era policy – like severity of lockdown, prevalence of parents' rights groups — on tension in local school board meetings

Data: We've been extended access to DistrictView, a corpus of school board meeting transcripts spanning 90% of districts from 2005 to 2025. That is our main source of data, but we also have data from Stanford on educational outcomes at the district level, from Emily Oster's COVID School Data Hub on school closures, and from Safegraph's cell-phone use data near school geographies.

References

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Data Sources

CALDER. (2024). *Working Paper 309.* https://caldercenter.org/sites/default/files/2024-11/CALDER-WP-309-1124.pdf

DistrictView. https://districtview.net/

Oster, E., et al. COVID School Data Hub. https://www.covidschooldatahub.com/

Open Science Framework project (Safegraph Cell Phone Data). https://osf.io/tpwqf/

Reardon, S. F., et al. Stanford Education Data Archive (SEDA). https://edopportunity.org